Effects of Academic and Career Factors on Early-Career Outcomes: A Categorical Data Analysis

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MATH 327: Categorical Data Analysis

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Objective

In this project, we will explore the relationship between academic and career factors like field of study, number of internships completed, and number of job offers to determine their effects on early-career outcomes. As soon-to-be college graduates, we wanted to get a better understanding on how academic and professional experiences can influence our careers and general life satisfaction post-grad. This categorical data analysis will provide us with the information we need to plan accordingly and make more informed decisions as we are about to enter the workforce.

The goal of this project is to examine how academic and career-related factors like field of study and number of internships are associated with early-career outcomes, including job offers and career satisfaction. Our modeling objective is inferential as we are just trying to understand the relationships among academic and career variables and not actually predict future outcomes. Three research questions were developed that stem from the overarching goal. (1) Is there an association between internships completed and job offers? (2) Is the proportion of high career satisfaction graduates significantly different between STEM and non-STEM fields? (3) Among field of study, internships, and job offers, which loglinear model (testing mutual independence, joint independence, marginal independence, conditional independence, and homogeneous association) best explains their interrelationships?

The goal of the first research question is to determine if the number of internships completed has a statistically significant association with the amount of job offers. The variable *Job Offers* was converted into a dummy variable to make it easier to work with in the analysis. A two-way contingency table was created with the counts for variable *Internships Completed* as the rows and the counts for variable *Job Offers* as the columns. A chi-square test of independence

was run with the categorical dependent variable being *Job Offers* and the categorical independent variable being *Internships Completed*. This test will show if the distribution of job offers differs significantly across the number of internships. The chi-square statistic, degrees of freedom, and corresponding p-value will be used to evaluate significance.

The goal of the second research question is to determine whether field of study (STEM vs. non-STEM) is associated with differences in self-reported high career satisfaction. The independent variable is *Field of Study* (grouped as STEM vs. non-STEM) and the dependent variable is *Career Satisfaction* (grouped as high vs. not high). A 2×2 contingency table was created to summarize the counts of graduates in each category. Proportions of graduates with high career satisfaction were then computed for each field. Finally, to test whether these proportions are significantly different, a two-proportion z-test was conducted.

The goal of the third research question is to determine which loglinear model best explains the relationships among variables *Field of Study*, *Internships Completed,* and *Job Offers*. To test this, a loglinear analysis was used in R. Several models including mutual independence, joint independence, marginal independence, conditional independence and homogeneous association were fitted. The chi-squared statistics and p-values for each model was compared to determine which model best fits the data.

Introduction

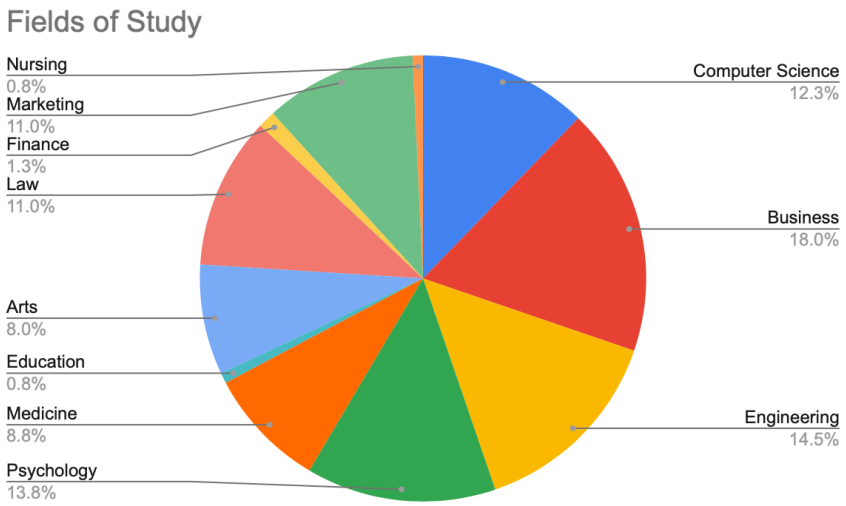
The "Education & Career Success" dataset from Kaggle provides a view into how students' educational achievements provide "real-world career outcomes." This dataset includes 400 records from students' academic performance, their extracurriculars, and their outcomes once they started their career. There are originally 19 different variables in this dataset, but only four variables were selected for this specific analysis due to their expected association.

The full dataset includes a wide range of variables, such as *Student ID*, *Age*, *Gender*, *High School GPA*, *SAT Score*, *University GPA*, *Field of Study*, *Internships Completed*, *Projects Completed*, *Certifications*, *Soft Skills Score*, *Networking Score*, *Job Offers*, *Starting Salary*, *Career Satisfaction*, *Years to Promotion*, *Current Job Level*, *Work Life Balance*, and *Entrepreneurship*. *Age*, *High School GPA*, *SAT Score*, *University GPA*, *Internships Completed*, *Projects Completed*, *Certifications*, *Soft Skills Score*, *Networking Score*, *Job Offers*, *Starting Salary*, *Career Satisfaction*, *Years to Promotion*, and *Work Life Balance* are all numerical variables. *Student ID*, *Gender*, *Field of Study*, *Current Job Level*, and *Entrepreneurship* are all categorical variables.

The dataset was cleaned to isolate necessary variables and prepare the data for analysis. The variables *Field of Study*, *Internships Completed*, *Jobs Offered*, and *Career Satisfaction* will be analyzed for this categorical data analysis. *Internships Completed* were grouped as 0 internships, 1 internship, 2 internships, and 3 or more internships to list it as a categorical variable. The same process was used for *Jobs Offered*, grouping them as: 0 *jobs offered*, 1 job offered, 2 *jobs offered*, 3 or more *jobs offered*. A combination of these variables will be used to test and analyze how academic experience relates to job outcomes. The variable *Field of Study* was grouped into two categories: “STEM” and “Non-STEM”. STEM fields were defined as computer science, engineering, and medicine. Non-STEM fields were defined as business, psychology, arts, law, marketing, finance, education, and nursing. Finally, the variable *Career Satisfaction* was grouped into two categories: “High score” and “Not high score”. A high score was defined as any score 6-10. A high score was defined as any score 1-5.

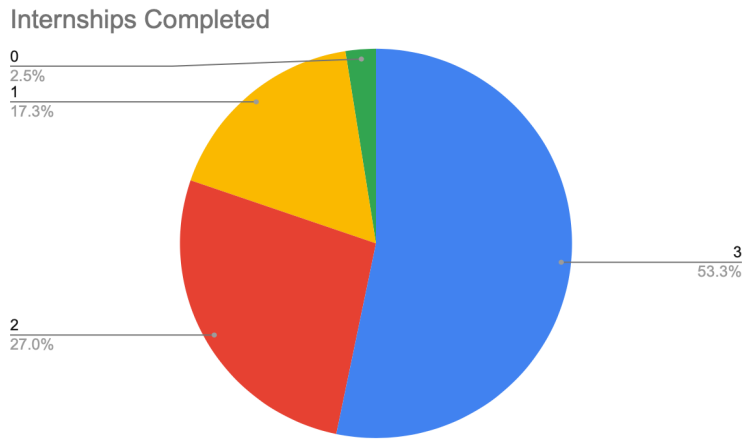
Field of Study is a categorical variable and it shows what each individual graduated with after their college education. It is a nominal variable in this dataset based on the variety of majors

individuals have in this dataset. It is an independent variable in this dataset and is a key part in determining the starting salary for the people in this study. It is an important factor because it is a gateway to what one might want to do throughout their life. The following pie chart visualizes how many people participated in each different field of study. The most common field of study is business with 18% of the individuals and nursing and education have the least common with 0.8% of the individuals.

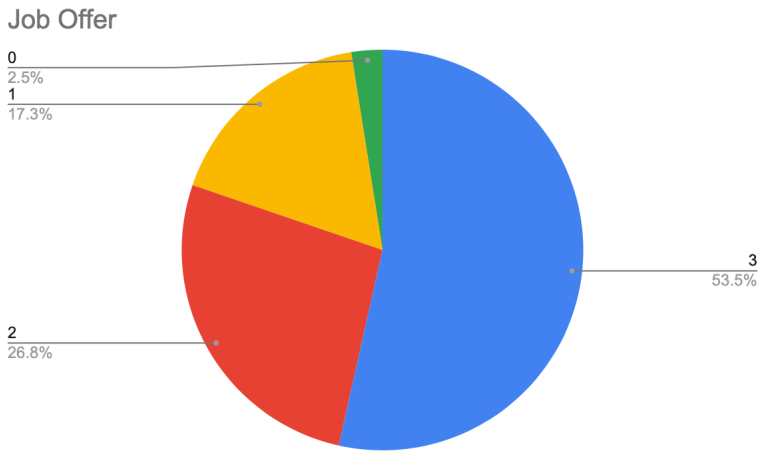


Internships Completed is a categorical variable and it shows which individuals completed one (1), two (2), multiple (3), or zero (0) internships. It is an ordinal categorical variable because the categories (0 internships, 1 internship, 2 internships, and 3 or more internships) have a natural ordering based on quantity. It is an independent variable in helping determine the number of job offers and whether the internships actually help in increasing the amount of job offers one receives. Internships are an important factor because they help guide someone into the career life and workforce through steps whether it's paid or volunteering. The following pie chart visualizes this variable and shows the distribution of those who have completed an internship, multiple

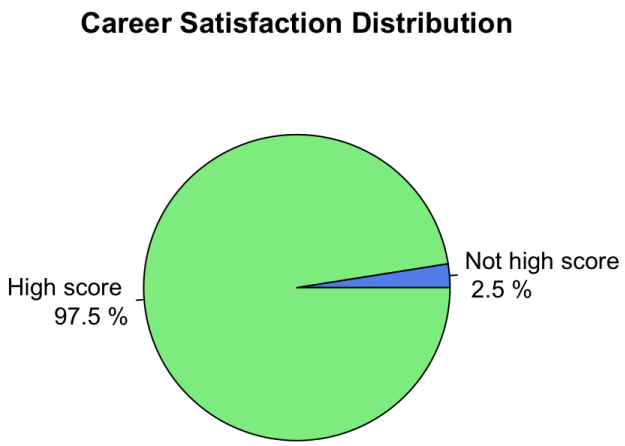
internships, or never completed an internship. Only 2.5% of individuals never completed an internship while over 50% of people in the study completed more than 2 internships.



Jobs Offered is a variable described as the "total number of distinct full-time job offers received within six months of graduation." These numbers were classified as 0 job offers, 1 job offer, 2 job offers, and three or more job offers. The variable *Jobs Offered* is an ordinal variable since it is a natural listed order. It is used as a categorical variable in the chi-square and loglinear analyses to examine whether it is associated with internships completed and how it interacts with variable *Internships Completed*. The amount of job offers is an important variable to include in the predictors because it usually indicates that the applicant or candidate has optimal skills for the specific career they are trying to achieve.



Career Satisfaction was originally a numerical variable ranging from a score of 1-10, with 1 representing low satisfaction and 10 representing high satisfaction. However, it was coded into two categories "High Score" (6-10) and "Not High Score" (1-5). It is an ordinal categorical variable because the two categories still represent a distinction between high satisfaction and low satisfaction. It is functioning as a dependent variable in this analysis because it represents an outcome influenced by the other variables mentioned above like field of study, internship experience, and job offers. The following pie chart shows the distribution of individuals across the two satisfaction groups. The majority of participants fall into the "High Score" category.



Analysis

Research Question #1: Is there an association between the amount of internships completed and job offers?

For this question we first had to find out the amount of people in the data set that had completed either zero internships, one, two, or three plus internships.

Job Offered	0 Internships	1 Internship Completed	2 Internship Completed	3+ Internship Completed
0 = No Job Offered	10	0	0	0
1 = Job Offered	0	69	108	213

In the table, we can see that we have 10 applicants with zero *internships completed* that received zero job offers. We can also observe that 69 applicants with one *internship completed* were offered a job, 108 applicants with 2 *internships completed* were offered a job, and those with 3 or more *internships completed* were offered a job. Now that we have a table that has our specific data and numerical values that we want to look at when answering this question, we can plug our numbers into R code and run some tests.

When looking for an association between two different variables, we can run a test of independence to see if the distribution of one variable differs, depending on the value of the other variable. If the variables are the same for all values of the other variable, we can conclude the variables would be independent. So we are going to run a Hypothesis Test of Independence to get a more clear understanding of the dataset. Our Null Hypothesis is going to be *Internships completed* and *Jobs offered* are independent, while our Alternate Hypothesis is going to be *Internships completed* and *Jobs Offered* are not independent.

```
> obs<-matrix(c(10,0,0,0,0,69,108,213),ncol = 4,byrow = TRUE)
> obs
     [,1] [,2] [,3] [,4]
[1,]   10    0    0    0
[2,]    0   69  108  213
> chisq.test(obs)

        Pearson's Chi-squared test

data:  obs
X-squared = 400, df = 3, p-value < 2.2e-16
```

After plugging in our values into the R Programming Software with the corresponding code, we receive the output that can be seen above. As we observe this output, we can see we were given a Test-Statistic (X^2) of 400, a Degrees of Freedom of 3, and a P-Value of < 2.2e-16. When we look at our P-Value, we get a number of approximately 0, so we can make the assumption that our P-Value < 0.05=α, so we Reject the Null Hypothesis. After rejecting our Null Hypothesis, *Internships completed* and *Jobs offered* are independent, we can conclude that *Internships completed* and *Jobs offered* are not independent of each other.

After receiving these results and making a strong conclusion about the Test of Independence with this specific dataset, we learn some important information about the overall idea with this test and specific dataset. The reason we had this question was to see the true importance of internships and how much of an impact they have on whether or not they are offered a job after the completion of internships. We can now see that a key aspect on whether or not you get offered a job after your education is through completion of internships, no matter if it's one, or multiple internships.

Research Question #2*:* Is the proportion of high career satisfaction graduates significantly different between STEM and non-STEM fields?

For this question, we first determined how many graduates in the dataset belonged to STEM fields and non-STEM fields, and within each group, how many reported high career satisfaction vs. not high career satisfaction. The field of study variable was cleaned and recategorized into two groups. STEM fields were defined as computer science, engineering and medicine, while non-STEM fields were defined as business, psychology, arts, law, marketing, finance, education and nursing. Career satisfaction was originally measured on a scale from 1 to 10. For this analysis, the variable was grouped into two categories: scores from 1 to 5 were

classified as not high career satisfaction, and scores from 6 to 10 were classified as high career satisfaction.

After organizing the data, we created a 2x2 contingency table that shows how many graduates in each field reported high career satisfaction versus those who did not. This table allows us to clearly compare the two groups and observe any possible differences in proportions.

```
> data <- read.csv("~/downloads/cleaned_education_data.csv")
> colnames(data)
[1] "Field_of_Study_Category"      "Internships_Category"
[3] "Job_Offers_Category"          "Career_Satisfaction_Category"
> table<- table(data$Field_of_Study_Category,
+                   data$Career_Satisfaction_Category)
> table

          High score Not high score
  Non-STEM        248             10
  STEM            142              0
```

We observed that all 142 STEM graduates reported high career satisfaction, while 248 out of 258 Non-STEM graduates reported high career satisfaction. This shows that although both groups have very high levels of career satisfaction, the proportion is slightly higher for STEM graduates.

Finally, we conducted a two-proportion z-test to test whether the difference between the two proportions is statistically significant. The test compared the proportion of graduates with high career satisfaction in the STEM group to that of the Non-STEM group.

```
> prop.test(
+   c(142, 248),
+   c(142, 258)
+ )

        2-sample test for equality of proportions with continuity correction

data:  c(142, 248) out of c(142, 258)
X-squared = 4.1668, df = 1, p-value = 0.04122
alternative hypothesis: two.sided
95 percent confidence interval:
 0.00974768 0.06777170
sample estimates:
   prop 1    prop 2
1.0000000 0.9612403
```

The results of the two-proportion test gave a p-value of 0.04122. Since this is less than the significance level of 0.05, we reject the null hypothesis. This provides statistically significant evidence that the proportion of graduates with high career satisfaction is different between STEM and Non-STEM fields.

In conclusion, although both STEM and Non-STEM graduates report very high levels of career satisfaction overall, the results suggest that STEM graduates have a slightly higher proportion of high career satisfaction compared to Non-STEM graduates, and this difference is statistically significant.

Research Question #3*:How does job level and gender relate to those with high career satisfaction?*

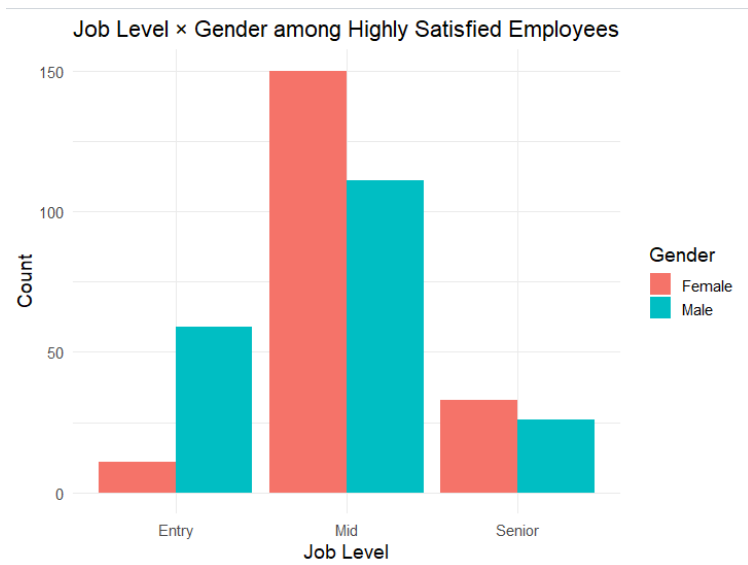
Career satisfaction is an important indicator of employee wellbeing and success in the workplace, it can influence factors such as how hard an employee works, how they view their job, and what the quality of their work is when presented. This research question aims to examine how the variables Gender and Job Level relate to having a high career satisfaction by testing how these factors interact with each other. By analyzing this data we can reveal which

groups experience the highest job satisfaction whether it be new employees, seniors within the company, males, or females.

We used both the chi-square test and Fisher test in order to test these interactions, the chi square was used in order to test interactions between categorical variables and the fishers test to get a more accurate reading due to sparse or small data. These tests allow us to see whether certain jobs have disproportionately skewed gender gap between high ranking levels of work and helps to confirm that the interactions in the data we are using are meaningful rather than random.

```
        Pearson's Chi-squared test

data:  table1
X-squared = 39.563, df = 2, p-value = 2.564e-09

> fisher.test(table1)

        Fisher's Exact Test for Count Data

data:  table1
p-value = 6.082e-10
alternative hypothesis: two.sided
```

Both tests gave us extremely low p values of 2.564 x 10e-9 and 6.082 x 10e-10, because of this, the null hypothesis of independence is rejected meaning that gender distribution is not equal among employees. For example, at entry level positions there are more men than women, while at mid positions females are higher than men. The chi-square and fisher tests confirm that these differences are statistically meaningful and not due to random chance.



In conclusion, there is statistical evidence that gender disparity exists in certain levels of jobs, indicating that men either get into or stay in entry positions more than women, and that women tend to work their way up and stay in mid level positions more than men.

Results and Limitations

In conclusion, our research and categorical data analysis uncovered how factors like field of study and internships completed interact and contribute to differences in early-career outcomes like jobs offered and career satisfaction. Our research showed that there is a strong association between the number of internships one completes and the likelihood of receiving a job offer. To put it simply, this means that completing more internships significantly increases the probability of receiving a job offer. Additionally, even though both STEM and Non-STEM groups reported very high overall career satisfaction, students who graduated with a degree in STEM fields were a little bit more likely to report high career satisfaction than students who graduated with a degree in a Non-STEM field. Since we wanted to see how gender and job level is influenced when dealing with a high level of career satisfaction. From the Chi-Square and Fisher test, these low p-values indicate that the distribution between genders differs for job level, while keeping the satisfaction score constant.

References

AdilShamim8. Education and Career Success. Kaggle, 2024. Web. 5 Dec. 2025. https://www.kaggle.com/datasets/adilshamim8/education-and-career-success.

R Script

```
#MATH 327 Final Project Report
data <- education_career_success
#PREPARE THE DATA
#Create Field of Study Category (STEM vs Non-STEM)
stem_fields <- c("computer science", "engineering", "medicine")
data$Field_of_Study_Category <- ifelse(
  tolower(data$Field_of_Study) %in% stem_fields,
  "STEM",
  "Non-STEM" )

#Create Internships Category
data$Internships_Category <- cut(data$Internships_Completed,
                          breaks = c(-1, 0, 1, 2, 100),
                          labels = c("0 internships", "1 internship",
                                  "2 internships", "3 or more internships"))

#Create Job Offers Category
data$Job_Offers_Category <- cut(data$Job_Offers,
                          breaks = c(-1, 0, 1, 2, 100),
                          labels = c("0 jobs offered", "1 job offered",
                                  "2 jobs offered", "3 or more jobs offered"))
#Create Career Satisfaction Category
data$Career_Satisfaction_Category <- cut(data$Career_Satisfaction,
                                 breaks = c(0, 5, 10),
                                 labels = c("Not high score", "High score"))

# Keep only the 4 variables we need
cleaned_data <- data[, c("Field_of_Study_Category",
                     "Internships_Category",
                     "Job_Offers_Category",
                     "Career_Satisfaction_Category")]
write.csv(cleaned_data, "cleaned_education_data.csv", row.names = FALSE)
 getwd()
[1] "/Users/spezzullo"
#RESEARCH QUESTION 1
Obs <- matrix(c(10,0,0,0,0,69,108,213),ncol = 4,byrow = TRUE)
Obs
     [,1] [,2] [,3] [,4]
[1,]   10    0    0    0
[2,]    0   69  108  213
chisq.test(Obs)

        Pearson's Chi-squared test
```

```
data:  Obs
X-squared = 400, df = 3, p-value < 2.2e-16

#RESEARCH QUESTION 2
> table<- table(data$Field_of_Study_Category,
+             data$Career_Satisfaction_Category)
> table

          Not high score High score
  Non-STEM             10        248
  STEM                  0        142
> prop.test(
+   c(142, 248),
+   c(142, 258)
+ )

        2-sample test for equality of proportions with continuity correction

data:  c(142, 248) out of c(142, 258)
X-squared = 4.1668, df = 1, p-value = 0.04122
alternative hypothesis: two.sided
95 percent confidence interval:
 0.00974768 0.06777170
sample estimates:
   prop 1    prop 2
1.0000000 0.9612403

Warning message:
In prop.test(c(142, 248), c(142, 258)) :
  Chi-squared approximation may be incorrect
```